INTA 4803 / 8803 - Space Sustainability

Notice to Lunar Operators

A NOTAM-Inspired Regime for Deconflicting Lunar Surface Operations Under High Cadence

moon.jpg

Clara Poitevin
Jalen Cauley
Dylan Gantt
Erik Goeke
Waleed Sadiq
Rut Santana

April 1, 2026

Georgia Institute of Technology

1 History of the NOTAM

NOTAMs were inspired by the Notice to Mariners (NtM), which has been created and distributed by the National Geospatial-Intelligence Agency (NGA) since 1886 [1]. The NtM was created because most mariners aboard ships have limited access to the internet and rely on paper charts for navigation. However, these charts may be outdated. To address this issue, the NGA compiles safety alerts and updates to navigational charts and publishes them in a weekly report [1]. This system served as the foundation for what would later become the NOTAM.

The Provisional International Civil Aviation Organization (PICAO) was established in 1945 to standardize aviation communication systems, including NOTAMs, which were formalized in 1947 [2]. Following the signing of the Chicago Convention in 1944, PICAO transitioned into the International Civil Aviation Organization (ICAO) [3]. Today, ICAO consists of 193 member states, strengthening its global authority in aviation standards [4].

NOTAMs are required to be reviewed by pilots prior to flight under 14 CFR 91.103, which states that pilots must familiarize themselves with all available flight information [5]. This includes NOTAMs. Additionally, NOTAMs are referenced in FAA publications related to both aviation and space operations [6].

31-3-2. NOTICE TO AIRMEN (NOTAM)

a. NOTAMs issued for space launch and reentry operations will be processed in accordance with current FAA directives.

b. The NOTAM must include the key words “airspace," “space launch," or “space reentry;" the launch or reentry site description, effective dates and times, and a chart depicting the area boundaries. It should also include a brief narrative describing the launch or reentry scenario, activities, types of launch or reentry vehicle involved, and the availability of inflight activity status information for nonparticipating pilots.

c. Information regarding the methods of airspace management may also be addressed.

Figure 1: Example of NOTAM usage in space-related operations

As shown above, even space launches require NOTAM dissemination to aircraft in nearby airspace. This concept directly inspired the development of the NOTLO.

2 NOTAM Problems and the Creation of the NOTLO

A primary design goal of the NOTLO was clarity and accessibility. Traditional NOTAMs have been widely criticized for being overly complex and difficult to interpret. Former National Transportation Safety Board chairman Robert Sumwalt described NOTAMs as “a bunch of garbage that no one pays any attention to” and suggested they are only understandable by computer programmers [7].

In response to these concerns, the NOTAM Improvement Act of 2023 was introduced, requiring the FAA to improve how NOTAM information is organized and presented to optimize pilot comprehension [8]. The FAA has also initiated modernization efforts, aiming to transition to a streamlined, single-source NOTAM system by 2026 [9].

NOTAMs also include specialized subtypes to address different hazards. For example, a SNOWTAM reports hazardous runway conditions due to snow or ice [10], while an ASHTAM provides information about volcanic activity and ash clouds that may impact aircraft operations [11]. Despite these expansions, complexity remains a major issue. Even attempts to rename NOTAMs to “Notices to Air Missions” for inclusivity were short-lived [12].

Building upon these limitations, the NOTLO (Notice to Lunar Operators) was developed as a more intuitive and accessible system.

2.1 DUSTLO Example

```
NOTLO 001/26 · TYPE: DUSTLO

DUSTLO 001/2026 — DUST HAZARD NOTICE  |  NASA
LOCATION:   Artemis Base, Pad Bravo  (00.00°S 023.47°E)
VALID:      21 APR 2026  14:00-16:00 UTC
VEHICLE:    ORION |  Departure heading 090°
HAZARD:     Dust plume 3 km radius
ACTIONS:    EVA/rover hold within 5 km until 30 min post-liftoff
            Pressurised suit mandatory outside habitats
            Deploy solar panel covers in dust path
AUTH: NASA-SCI-001
```

Figure 2: Example of a DUSTLO notification

The DUSTLO is issued during lunar takeoff or landing operations, where regolith dispersion poses a hazard. In this example, a NASA Orion spacecraft launches from Artemis Base, Pad Bravo, heading east. A 3 km dust plume is expected, requiring all EVA and rover activity to cease within a 5 km radius. Operations may resume 30 minutes after liftoff. Additional precautions include wearing pressurized suits and covering solar panels.

2.2 RADLO Example

```
NOTLO 002/26 · TYPE: RADLO

RADLO 007/2026 — RADIATION EXPERIMENT NOTICE  |  NASA
LOCATION:   Shackleton Crater Rim, Sector 7  (89.92°S 002.14°E)
VALID:      21 APR - 28 APR 2026  (continuous)
EXPERIMENT: Cosmic Ray Detector Array
TRANSMIT:  4 GHz |  Comms degraded within 500 m
ACTIONS:    200 m personnel exclusion at all times
            50 m hard exclusion (no metal equipment)
            Reroute all rover traverses around Sector 7
AUTH: NASA-SCI-002
```

Figure 3: Example of a RADLO notification

The RADLO is used for radiation-related hazards. In this case, a cosmic ray detector array emits radiation at specific frequencies. The RADLO specifies affected frequencies, operational timing, and exclusion zones to ensure astronaut and rover safety.

2.3 EXPLO Example

```
NOTLO 003/26 · TYPE: EXPLO

EXPLO 012/2026 — ACTIVE EXPERIMENT NOTICE  |  NASA
LOCATION:   Erik Base, Science Sector 12  (00.67°N 023.43°E)
CAMPAIGN:   21 APR – 28 APR 2026  |  Active nightly 18:00–06:00 UTC
EXPERIMENT: Passive Seismometer Array — moonquake detection
ACTIONS:    300 m quiet zone — no EVA, rovers, or equipment movement
            Zone reopens daily at 06:00 UTC
AUTH: NASA-SCI-003
```

Figure 4: Example of an EXPLO notification

The EXPLO applies to sensitive experiments or restricted operational zones. In this example, a seismic experiment requires minimal surface disturbance. As a result, rover and EVA activity are restricted during operational hours, which occur in the evening.

Each NOTLO includes timing, hazard descriptions, exclusion zones, and contact information, ensuring clarity and usability.

2.4 Design Philosophy of the NOTLO

The NOTLO system prioritizes simplicity, readability, and accessibility. Unlike traditional NOTAMs, which can be difficult for non-specialists to interpret, NOTLOs are designed to be understood by a broad audience, including the general public. Another main design goal for the NOTLO was for it to be easily machine readable, this will make it again so that tools can be developed for the NOTLO database in order to solve any future problems that may occur with the system. If a consistent data structure is used and saved in a database, it would also be very helpful in the programming of future automated missions. This system would also be queryable by a variety of filters such as mission type, mission duration, and mission dates, this will make it so that analysis can be done simply and quickly, without the need for extensive searching. NOTAMs also often suffer from inconsistent phrasing such as ”out of service”, taking this as an initiative to improve, the NOTLO will keep inconsistent wording out of its announcements. Furthermore, in order not to leave the lunar environment in disarray with a large magnitude of announcements, there will be a filtration system, and announcements will be sent out with a higher priority score to those lunar players who it impacts more.

Additionally, the NOTLO framework supports a centralized database for both real-time and historical notices. This ensures transparency and long-term usability, addressing a key limitation of the current NOTAM system. Establishing a clear and scalable structure early in development is critical to ensuring the system remains effective as lunar operations expand, and will also ensure that there is a distribution architecture present when the lunar environment is more saturated. Finally, in order to ensure that NOTLOs may still be sent out in the event of a communications outage, there will be storage systems, and dedicated data centers on the moon used in the distribution of NOTLOs. this will ensure that even if communication is cut off with Earth, NOTLOs can still be sent out and serve their purpose of keeping the lunar environment safe.

References

[1] U.S. Navy DON CIO. *NOTAM-related Information Article*. Accessed: 2026-04-25. n.d. URL: https://www.doncio.navy.mil/chips/ArticleDetails.aspx?ID=12945.

[2] AirSupport. *The Evolution of NOTAMs*. Accessed: 2026-04-25. n.d. URL: https://airsupport.website/the-evolution-of-notams/.

[3] Federal Aviation Administration. *Pioneers of ICAO (PICAO)*. Accessed: 2026-04-25. n.d. URL: https://www.faa.gov/sites/faa.gov/files/about/history/pioneers/PICAO.pdf.

[4] International Civil Aviation Organization. *ICAO Member States*. Accessed: 2026-04-25. n.d. URL: https://www.icao.int/about-icao/member-states.

[5] Electronic Code of Federal Regulations. *14 CFR § 91.103 - Preflight Action*. Accessed: 2026-04-25. n.d. URL: https://www.ecfr.gov/current/title-14/chapter-I/subchapter-F/part-91/subpart-B/subject-group-ECFRe4c59b5f5506932/section-91.103.

[6] Federal Aviation Administration. *Pilot/Controller Glossary - NOTAM*. Accessed: 2026-04-25. n.d. URL: https://www.faa.gov/air_traffic/publications/atpubs/pham_html/chap31_section_3.html#31-3-2.

[7] Chad Trautvetter. "NTSB Chairman Calls NOTAMs "Garbage"". In: *Aviation International News* (Sept. 2018). Accessed: 2026-04-25. URL: https://www.ainonline.com/aviation-news/business-aviation/2018-09-28/ntsb-chairman-calls-notams-garbage.

[8] U.S. Congress. *H.R.346 - NOTAM Improvement Act of 2023*. Public Law No: 118-4, Accessed: 2026-04-25. 2023. URL: https://www.congress.gov/bill/118th-congress/house-bill/346.

[9] Federal Aviation Administration. *What is a NOTAM?* Accessed: 2026-04-25. n.d. URL: https://www.faa.gov/about/initiatives/notam/what_is_a_notam.

[10] Skybrary. *SNOWTAM*. Accessed: 2026-04-25. n.d. URL: https://skybrary.aero/articles/snowtam.

[11] Skybrary. *ASHTAM*. Accessed: 2026-04-25. n.d. URL: https://skybrary.aero/articles/ashtam.

[12] Pilot Institute. *What Are NOTAMs? Notices to Air Missions Explained*. Accessed: 2026-04-25. n.d. URL: https://pilotinstitute.com/what-are-notams-notices-to-air-missions-explained/.